The topic is

- The topic is a general term that refers to the subject or main idea of a text, speech, or conversation.
- The topic can be expressed in different ways, such as a word, a phrase, a question, or a theme.
- The topic can be identified by looking for clues in the text, such as the title, the introduction, the headings, the keywords, or the summary.
- The topic can be used to organize and structure the text, as well as to guide the reader's or listener's attention and comprehension.
- The topic can be related to other topics, such as subtopics, supporting details, examples, or arguments, that help to develop and explain the main idea.
- The topic can vary in scope and complexity, depending on the purpose, audience, and genre of the text, speech, or conversation.

Workshop Practice Lab:

- Workshop practice lab is a common theory/lab course for all engineering students in the first and second semester .
- The objective of this lab is to get a hands-on knowledge of several workshop practices like carpentry, fitting, welding, machining, etc. and learn safety regulations to be maintained in a shop floor.
- Workshop practice is the backbone of the real industrial environment which helps to develop and enhance relevant technical hand skills required by the technician working in the various engineering industries and workshops.
- The lab syllabus may vary depending on the university or college, but some of the common experiments are :
  - Introduction to workshop tools and safety precautions.
  - Carpentry: making various types of joints, patterns, and models using wood and plywood.
  - Fitting: making various types of joints, keys, and couplings using metal pieces and files.
  - Welding: performing various types of welding operations such as gas welding, arc welding, spot welding, etc. using different electrodes and metals.
  - Machining: operating various types of machines such as lathe, drilling, milling, grinding, etc. and making different shapes and profiles using cutting tools and workpieces.
  - Sheet metal: making various types of objects such as trays, boxes, funnels, etc. using sheet metal and bending, cutting, and joining techniques.
  - Plumbing: making various types of pipe joints, valves, and fittings using PVC pipes and fittings.

    - Electrical: performing various types of electrical wiring, soldering, and testing using electrical components and instruments.
- The lab work consists of four parts :
    - A pre-lab homework that the students must complete before coming to the lab.
    - The lab itself, answering all the warm-ups and predictions, and attaching data, results, graphs, and analysis.
    - A post-lab report that the students must submit after the lab, summarizing the objectives, procedures, observations, calculations, and conclusions of the experiment.
    - A viva-voce or oral examination that the students must face to demonstrate their understanding and skills of the experiment.
- The lab assessment is based on the following criteria :
    - Attendance and participation in the lab sessions.
    - Quality and accuracy of the pre-lab homework, lab work, and post-lab report.
    - Performance and confidence in the viva-voce or oral examination.
    - Adherence to the safety rules and regulations in the workshop.

S. No. Mechanical Workshop Duration

- S. No. stands for serial number, which is a way of numbering items in a list or a table.
- Mechanical workshop is a place where various mechanical processes and operations are performed, such as machining, welding, fitting, carpentry, etc.
- Duration is the amount of time that something lasts or takes to complete, such as a workshop session, a project, or a course.
- S. No. Mechanical Workshop Duration is a possible title or heading for a table that shows the serial number, the name, and the duration of different mechanical workshops offered by an institution or an organization.
- An example of such a table is:

S. No.	Mechanical Workshop	Duration
1	Basic Machining	2 hours
2	Advanced Welding	3 hours
3	Fitting and Assembly	4 hours
4	Carpentry Skills	2 hours
5	Hydraulic Systems	3 hours

1 Introduction to Mechanical workshop material, tools and machines 3 Hrs

- Mechanical workshop is a place where various types of materials, tools and machines are used to perform different operations such as cutting, shaping,

drilling, welding, etc.
- The main objectives of mechanical workshop are to provide practical knowledge and skills to the students, to familiarize them with the common tools and machines used in engineering, and to develop their creativity and problem-solving abilities.
- The basic materials used in mechanical workshop are metals, alloys, plastics, wood, etc. Each material has its own properties, advantages and disadvantages, and applications.
- The common tools used in mechanical workshop are classified into two categories: hand tools and power tools. Hand tools are operated manually by human force, such as hammers, screwdrivers, files, etc. Power tools are operated by electric or pneumatic motors, such as drills, grinders, saws, etc.
- The common machines used in mechanical workshop are classified into two categories: conventional machines and CNC machines. Conventional machines are operated by manual controls, such as lathes, milling machines, shapers, etc. CNC machines are operated by computerized controls, such as CNC lathes, CNC milling machines, CNC routers, etc.
- The safety rules and precautions are very important in mechanical workshop, as there are many hazards and risks involved in working with materials, tools and machines. Some of the general safety rules are: wear appropriate personal protective equipment, follow the instructions and guidelines, keep the work area clean and organized, use the right tool or machine for the right job, report any accidents or defects, etc.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some content in markdown format that you can use to study layout, safety measures and different engineering materials used in workshop.

Layout, Safety Measures and Different Engineering Materials Used in Workshop

Layout of Workshop

- The layout of a workshop is the arrangement of machines, equipment and workstations in a way that optimizes the workflow, safety and productivity of the workers.
- The layout of a workshop depends on factors such as the type and size of the workshop, the nature and volume of the work, the available space and resources, the environmental conditions and the ergonomic aspects.
- The layout of a workshop can be classified into four types: process layout, product layout, cellular layout and fixed position layout.
- Process layout: In this type, machines and equipment are grouped according to their function or process, such as drilling, milling, welding, etc. This allows for flexibility and variety of work, but may result in longer travel

distances, higher material handling costs and lower efficiency.

- Product layout: In this type, machines and equipment are arranged in a sequence that follows the production process of a specific product or a family of products. This allows for high efficiency, low material handling costs and shorter production time, but may result in less flexibility and higher initial investment.
- Cellular layout: In this type, machines and equipment are grouped into cells that produce a part or a subassembly of a product. Each cell is designed to operate as a self-contained unit, with minimal dependence on other cells. This allows for high flexibility, low material handling costs and improved quality, but may result in higher space requirements and complex scheduling.
- Fixed position layout: In this type, the product remains in a fixed position and the machines and equipment are brought to it as needed. This is suitable for large, bulky or complex products that are difficult to move, such as ships, aircraft, bridges, etc. This allows for high customization, low material handling costs and reduced work-in-process inventory, but may result in low efficiency, high space requirements and coordination problems.

Safety Measures in Workshop

- Safety measures in workshop are the rules, regulations and precautions that are followed to prevent accidents, injuries and hazards in the workshop environment.
- Safety measures in workshop include the following aspects:
  - Personal protective equipment (PPE): These are the devices and garments that are worn by the workers to protect them from the risks of exposure to harmful substances, noise, heat, radiation, etc. Examples of PPE are gloves, goggles, helmets, masks, ear plugs, aprons, etc.
  - Machine guarding: These are the barriers and covers that are installed on the machines and equipment to prevent the workers from coming into contact with the moving parts, sharp edges, sparks, etc. Examples of machine guarding are guards, shields, fences, interlocks, etc.
  - Fire prevention and protection: These are the measures that are taken to prevent and control the outbreak of fire in the workshop. Examples of fire prevention and protection are fire extinguishers, fire alarms, fire hoses, sprinklers, etc.
  - Electrical safety: These are the measures that are taken to prevent and avoid the hazards of electric shock, electrocution, short circuit, etc. in the workshop. Examples of electrical safety are grounding, insulation, circuit breakers, fuses, etc.
  - Housekeeping: These are the practices that are followed to keep the workshop clean, tidy and organized. Examples of housekeeping are

sweeping, dusting, disposing of waste, storing tools and materials, etc.
  - Ergonomics: These are the principles that are applied to design the machines, equipment and workstations in a way that reduces the physical and mental stress and fatigue of the workers. Examples of ergonomics are adjustable chairs, proper lighting, adequate ventilation, etc.

Different Engineering Materials Used in Workshop

- Engineering materials are the substances that are used to make or modify the products or components in the workshop. Engineering materials can be classified into four categories: metals, ceramics, polymers and composites.
- Metals: These are the materials that are composed of one or more metallic elements, such as iron, copper, aluminum, etc. Metals have high strength, ductility, conductivity and resistance to corrosion, but may have low stiffness, brittleness and thermal expansion. Metals can be further classified into ferrous and non-ferrous metals, based on the presence or absence of iron in their composition.
  - Ferrous metals: These are the metals that contain iron as the main element, such as steel, cast iron, etc. Ferrous metals have high strength, hardness and durability, but may have low resistance to corrosion and oxidation. Ferrous metals can be further classified into mild steel, medium

To study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section.

- Fitting is the process of assembling or joining two or more parts by cutting, filing, drilling, tapping, etc. to make them fit together accurately and securely.
- Sheet metal is the process of forming thin sheets of metal into various shapes and sizes by cutting, bending, punching, riveting, etc.
- Welding is the process of joining two or more metal parts by applying heat, pressure, or both, with or without filler material, to form a strong and permanent bond.

Some of the common tools, equipment, devices and machines used in these processes are:

- Fitting tools: These include measuring tools (such as steel rule, caliper, micrometer, etc.), marking tools (such as scriber, punch, divider, etc.), cutting tools (such as hacksaw, chisel, file, etc.), drilling tools (such as drill bit, drill chuck, drill press, etc.), tapping tools (such as tap, tap wrench, tap holder, etc.), and fastening tools (such as hammer, screwdriver, spanner, etc.).

- Sheet metal tools: These include measuring tools (such as steel rule, caliper, micrometer, etc.), marking tools (such as scriber, punch, divider, etc.), cutting tools (such as shears, snips, nibbler, etc.), bending tools (such as bending brake, folding machine, etc.), punching tools (such as punch, die, press, etc.), riveting tools (such as rivet, rivet set, rivet gun, etc.), and soldering tools (such as solder, flux, soldering iron, etc.).
- Welding tools: These include measuring tools (such as steel rule, caliper, micrometer, etc.), marking tools (such as scriber, punch, divider, etc.), cutting tools (such as oxy-acetylene torch, plasma cutter, etc.), welding machines (such as arc welder, gas welder, spot welder, etc.), welding electrodes (such as metal rod, wire, flux, etc.), welding accessories (such as welding helmet, gloves, apron, etc.), and welding inspection tools (such as magnifying glass, dye penetrant, etc.).

To determine the least count of Vernier calliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments.

- The least count of an instrument is the smallest measurement that can be taken with it. It is equal to the ratio of the main scale division to the number of divisions on the vernier scale or the micrometer scale.
- The least count of a vernier calliper is given by:
  - Least count = 1 main scale division - 1 vernier scale division
  - Least count = main scale division / number of vernier scale divisions
- The least count of a vernier height gauge is the same as that of a vernier calliper, since both have a main scale and a vernier scale.
- The least count of a micrometer (screw gauge) is given by:
  - Least count = pitch of the screw / number of divisions on the thimble
  - Pitch of the screw is the distance moved by the spindle per revolution of the thimble.
- To take different readings over given metallic pieces using these instruments, the following steps are followed:
  - Vernier calliper:
    * Place the metallic piece between the jaws of the calliper and close them gently.
    * Read the main scale division that coincides with the zero of the vernier scale. This is the main scale reading (MSR).

    - Find the vernier scale division that coincides with any main scale division. This is the vernier scale reading (VSR).
    - The total reading is given by:
      - Total reading = MSR + (VSR x least count)
  - Vernier height gauge:
    - Place the metallic piece on a flat surface and adjust the height of the gauge until it touches the top of the piece.
    - Read the main scale division that coincides with the zero of the vernier scale. This is the main scale reading (MSR).
    - Find the vernier scale division that coincides with any main scale division. This is the vernier scale reading (VSR).
    - The total reading is given by:
      - Total reading = MSR + (VSR x least count)
  - Micrometer (screw gauge):
    - Place the metallic piece between the anvil and the spindle of the micrometer and rotate the thimble until it touches the piece gently.
    - Read the main scale division that coincides with the edge of the thimble. This is the main scale reading (MSR).
    - Read the division on the thimble that coincides with the datum line on the sleeve. This is the thimble reading (TR).
    - The total reading is given by:
      - Total reading = MSR + (TR x least count)

Machine Shop

- A machine shop is a place where various machines and tools are used to produce or modify parts and components of different materials, such as metal, wood, plastic, etc.
- Machine shops can perform various operations, such as cutting, drilling, milling, turning, grinding, welding, etc., depending on the type and capacity of the machines and tools available.
- Machine shops can be classified into different types, such as general-purpose, specialized, production, or prototype, depending on the nature and volume of the work they perform.
- Machine shops require skilled workers, such as machinists, operators, programmers, and inspectors, who can operate, maintain, and control the machines and tools, as well as follow the safety and quality standards.

- Machine shops also require proper planning, layout, organization, and management, to ensure the efficient and effective use of the resources, such as space, power, materials, time, and labor.
- Machine shops are essential for various industries, such as manufacturing, engineering, construction, automotive, aerospace, etc., as they provide the means to create or modify the parts and components that are needed for various products and systems.

Demonstration of working, construction and accessories for Lathe machine Perform operations on Lathe - Facing, Plane Turning , step turning, taper turning, threading, knurling and parting.

- A lathe machine is a machine tool that uses a cutting tool for removing the material from the surface of the workpiece which is held in the chuck for holding the workpiece and feed was provided by the tool on to the workpiece for the removal of material.
- The main parts of a lathe machine are headstock, tailstock, bed, carriage, cross slide, compound rest, tool post, apron, lead screw, feed rod, chuck, spindle, etc .
- The main accessories of a lathe machine are face plate, catch plate or dog plate, mandrel, steady rest, follower rest, drive plate, lathe carrier or lathe dog, three jaw chuck, four jaw chuck, collet chuck, etc .
- The main operations of a lathe machine are facing, plane turning, step turning, taper turning, threading, knurling and parting.
- Facing is the operation of machining the ends of a workpiece to produce a flat surface square with the axis.
- Plane turning is the operation of reducing the diameter of a cylindrical workpiece over a length.
- Step turning is the operation of producing different steps of different diameters on a workpiece.
- Taper turning is the operation of producing a conical surface by gradually reducing the diameter of the workpiece along its length.
- Threading is the operation of cutting a helical groove on a cylindrical or conical surface of the workpiece.
- Knurling is the operation of producing a regular cross-hatched pattern on the surface of the workpiece for better grip or appearance.
- Parting is the operation of cutting off a piece of the workpiece from the main part.
- The following diagram shows the basic lathe machine and its parts:

Lathe machine and its parts

- The following diagram shows the basic lathe machine operations:

Lathe machine operations

- The following diagram shows some of the common lathe machine accessories:

Lathe machine accessories

Fitting Shop

- Fitting shop is a place where metal components are fitted together by filing, chiseling, sawing, drilling, tapping, etc.
- Fitting shop is also used for repairing or modifying existing parts or assemblies.
- Fitting shop requires various tools and equipment, such as vices, hammers, files, hacksaws, drills, taps, dies, gauges, etc.
- Fitting shop operations can be classified into four categories: measuring, marking, cutting, and joining.
- Measuring is the process of determining the dimensions and tolerances of the workpiece or the finished product.
- Marking is the process of transferring the measurements and layout to the workpiece using marking tools, such as scriber, punch, divider, etc.
- Cutting is the process of removing excess material from the workpiece using cutting tools, such as chisel, hacksaw, file, drill, etc.
- Joining is the process of connecting two or more workpieces using fasteners, such as screws, bolts, nuts, rivets, etc. or by welding, brazing, soldering, etc.

1. Practice marking operations.

- Marking operations are the process of applying marks or symbols to a surface, such as paper, metal, wood, or fabric, for identification, measurement, decoration, or communication purposes.
- Marking operations can be performed by various methods, such as writing, drawing, stamping, engraving, printing, painting, or spraying.
- Marking operations can be classified into two types: permanent and temporary. Permanent marking operations create marks that are difficult or impossible to remove, while temporary marking operations create marks that can be easily erased or washed off.
- Marking operations can have different functions, such as:
  - Indicating the name, origin, ownership, or quality of a product or material.
  - Providing instructions, warnings, or information for users or consumers.
  - Enhancing the appearance, value, or appeal of a product or material.
  - Expressing personal or artistic creativity or style.
- Marking operations can be practiced by using various tools, materials, and techniques, such as:
  - Pens, pencils, markers, crayons, or chalk for writing or drawing on paper, cardboard, or other surfaces.
  - Stamps, stickers, labels, or tags for applying pre-made marks or sym-

bols on paper, plastic, or other surfaces.
    - Knives, scissors, punches, or cutters for cutting out shapes or patterns on paper, fabric, or other materials.
    - Hammers, chisels, drills, or saws for carving or engraving marks or symbols on wood, metal, stone, or other hard materials.
    - Ink, paint, dye, or pigment for printing, painting, or spraying marks or symbols on paper, fabric, metal, or other surfaces.
    - Needles, threads, yarns, or fabrics for sewing, embroidering, or knitting marks or symbols on cloth or other textiles.
    - Heat, electricity, light, or chemicals for burning, etching, or staining marks or symbols on paper, metal, glass, or other materials.

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2. Preparation of U or V -Shape Male Female Work piece which contains: Filing, Sawing, Drilling, Grinding.

- A U or V -shape male female work piece is a pair of metal pieces that fit together in a U or V shape, with one piece having a protruding part and the other having a corresponding recess.
- The purpose of this work piece is to demonstrate the skills of filing, sawing, drilling, and grinding metal materials using appropriate tools and machines.
- The steps for preparing this work piece are as follows:

Filing

- Filing is the process of removing excess material from a metal surface using a file, which is a metal tool with teeth that cut into the metal when moved back and forth.
- Filing is used to smooth, shape, and finish the metal surfaces, as well as to remove burrs, nicks, and scratches.
- To file a metal surface, the following steps are followed:
    - Secure the metal piece in a vise or clamp to prevent it from moving.
    - Select a suitable file for the desired shape and finish. For example, a flat file for a flat surface, a half-round file for a curved surface, a fine file for a smooth finish, etc.
    - Hold the file with both hands, one near the handle and one near the tip, and apply moderate pressure on the forward stroke, while lifting the file slightly on the backward stroke to avoid dulling the teeth.
    - Move the file across the metal surface in a diagonal direction, creating a cross-hatched pattern of fine grooves.

  - Check the progress of the filing by wiping the metal surface with a cloth or a brush, and use a steel rule or a caliper to measure the dimensions and angles.
  - Repeat the filing until the desired shape and finish are achieved.

Sawing

- Sawing is the process of cutting a metal piece into smaller pieces using a saw, which is a metal blade with teeth that cut into the metal when moved back and forth.
- Sawing is used to separate, shape, and reduce the size of the metal pieces, as well as to create slots, notches, and holes.
- To saw a metal piece, the following steps are followed:
  - Secure the metal piece in a vise or clamp to prevent it from moving.
  - Select a suitable saw for the desired cut. For example, a hacksaw for a straight cut, a coping saw for a curved cut, a fine-toothed saw for a thin metal, etc.
  - Mark the line of the cut on the metal surface using a scriber, a ruler, and a square.
  - Hold the saw with one hand on the handle and the other on the frame, and position the blade perpendicular to the metal surface, with the teeth pointing away from the handle.
  - Start the cut by making a few light strokes with the saw, creating a groove on the metal surface.
  - Continue the cut by applying moderate pressure on the forward stroke, while lifting the saw slightly on the backward stroke to avoid binding the blade.
  - Check the progress of the sawing by measuring the length and angle of the cut, and adjust the direction of the saw as needed.
  - Repeat the sawing until the metal piece is cut through.

Drilling

- Drilling is the process of creating holes in a metal piece using a drill, which is a metal tool with a pointed tip that rotates and cuts into the metal when pressed against it.
- Drilling is used to create holes for fastening, joining, or inserting other metal pieces, as well as to create recesses, countersinks, and counterbores.
- To drill a hole in a metal piece, the following steps are followed:
  - Secure the metal piece in a vise or clamp to prevent it from moving.
  - Select a suitable drill for the desired hole size and depth. For example, a twist drill for a round hole, a spade drill for a flat-bottomed hole, a step drill for a tapered hole, etc.
  - Mark the center of the hole on the metal surface using a center punch and a hammer, creating a small indentation that guides the drill.
  - Hold the drill with one hand on the handle and the other on the

  chuck, and position the tip of the drill on the center of the hole, perpendicular to the metal surface.
  - Start the drill by pressing the trigger or the switch, and apply moderate pressure on the drill, while keeping it steady and straight.
  - Continue the drilling by feeding the drill into the metal piece, while clearing the chips and

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have asked me to write on the topic of 4 Carpentry Shop 3 Hrs. Here is the content I have generated for you in markdown format:

4 Carpentry Shop 3 Hrs

- Carpentry is the skill of making and repairing wooden objects and structures, such as furniture, doors, windows, stairs, etc.
- A carpentry shop is a place where carpenters work with various tools and materials to create and fix wooden items.
- A carpentry shop can be divided into different sections, such as:
  - The cutting section, where wood is cut into different shapes and sizes using saws, chisels, planes, etc.
  - The assembly section, where wood pieces are joined together using nails, screws, glue, etc.
  - The finishing section, where wood surfaces are smoothed, polished, painted, stained, etc.
  - The storage section, where wood and other materials are kept and organized.
- A carpentry shop requires some basic equipment and safety measures, such as:
  - A workbench, where carpenters can place and work on their projects.
  - A vise, which is a device that holds wood firmly in place while working on it.
  - A hammer, which is a tool that is used to drive nails, pull nails, or strike other tools.
  - A screwdriver, which is a tool that is used to turn screws or bolts.
  - A drill, which is a tool that is used to make holes in wood or other materials.
  - A level, which is a tool that is used to check if a surface is horizontal or vertical.
  - A tape measure, which is a tool that is used to measure length, width, or height of objects or spaces.
  - A square, which is a tool that is used to check if angles are right angles or not.
  - A saw, which is a tool that is used to cut wood or other materials.
  - A plane, which is a tool that is used to shave off thin layers of wood to make it smooth or flat.

  - A chisel, which is a tool that is used to carve or shape wood by removing small pieces of it.
  - A sandpaper, which is a material that is used to smooth or polish wood surfaces by rubbing it against them.
  - A paintbrush, which is a tool that is used to apply paint, stain, or varnish to wood surfaces.
  - A safety goggles, which are a protective eyewear that is used to prevent wood chips, dust, or other particles from entering the eyes.
  - A gloves, which are a protective handwear that is used to prevent cuts, splinters, or burns from wood or tools.
  - A apron, which is a protective clothing that is used to prevent wood or paint stains from getting on the clothes.
- A carpentry shop can offer various services and products, such as:
  - Custom-made furniture, such as tables, chairs, cabinets, shelves, etc.
  - Repair and restoration of old or damaged furniture, such as fixing broken legs, hinges, drawers, etc.
  - Installation and maintenance of doors, windows, stairs, railings, etc.
  - Carpentry classes and workshops, where people can learn the basics of carpentry or improve their skills.

Study of Carpentry Tools, Equipment and different joints. Making of Cross Half lap joint, Half lap Dovetail joint and Mortise Tenon Joint

Carpentry is the art and skill of working with wood to create various structures, furniture, and objects. Carpentry requires the use of various tools and equipment to measure, cut, shape, join, and finish wood. Some of the common carpentry tools and equipment are:

- **Hammer**: A tool used to drive nails, pins, or other fasteners into wood. Hammers have a metal head and a wooden or rubber handle. There are different types of hammers, such as claw hammer, ball-peen hammer, sledgehammer, etc.
- **Saw**: A tool used to cut wood into different shapes and sizes. Saws have a blade with teeth that move back and forth or in a circular motion. There are different types of saws, such as hand saw, circular saw, jigsaw, coping saw, etc.
- **Chisel**: A tool used to carve, shape, or remove wood. Chisels have a sharp metal edge and a wooden or plastic handle. There are different types of chisels, such as bevel-edge chisel, mortise chisel, dovetail chisel, etc.
- **Plane**: A tool used to smooth, level, or shape wood. Planes have a blade that shaves off thin layers of wood as it is pushed or pulled along the surface. There are different types of planes, such as jack plane, smoothing plane, block plane, etc.
- **Drill**: A tool used to make holes in wood. Drills have a rotating bit that can be changed to suit different sizes and shapes of holes. There are

different types of drills, such as hand drill, electric drill, cordless drill, etc.

- **Screwdriver**: A tool used to drive screws into or out of wood. Screwdrivers have a metal tip that fits into the head of a screw and a handle that can be turned to rotate the tip. There are different types of screwdrivers, such as flat-head screwdriver, Phillips screwdriver, Torx screwdriver, etc.
- **Clamp**: A device used to hold two or more pieces of wood together while they are being glued, nailed, or screwed. Clamps have two jaws that can be adjusted to fit different thicknesses of wood and a mechanism that applies pressure to keep the jaws closed. There are different types of clamps, such as C-clamp, bar clamp, pipe clamp, etc.
- **Square**: A tool used to measure and mark right angles on wood. Squares have two arms that form a 90-degree angle and a scale that shows the length of each arm. There are different types of squares, such as try square, combination square, speed square, etc.
- **Level**: A tool used to check if a surface is horizontal or vertical. Levels have a bubble that moves inside a liquid-filled tube and indicates if the surface is level or not. There are different types of levels, such as spirit level, laser level, digital level, etc.
- **Tape measure**: A tool used to measure the length, width, or height of wood or other objects. Tape measures have a flexible metal or plastic tape that can be extended or retracted and a scale that shows the measurement in inches or centimeters. There are different types of tape measures, such as retractable tape measure, folding tape measure, steel tape measure, etc.

Carpentry also involves the use of different joints to connect two or more pieces of wood together. Joints can be classified into two main categories: **fastened joints** and **glued joints**. Fastened joints use nails, screws, dowels, biscuits, or other metal fasteners to hold the wood together. Glued joints use wood glue or other adhesives to bond the wood together. Some of the common carpentry joints are:

- **Butt joint**: A joint where the end of one piece of wood is joined to the edge or face of another piece of wood. This is the simplest and weakest type of joint and usually requires fasteners or glue to reinforce it.
- **Miter joint**: A joint where the ends of two pieces of wood are cut at an angle and joined together to form a corner. This is a common type of joint for making frames, boxes, or cabinets and can be reinforced with fasteners or glue.
- **Dado joint**: A joint where a groove is cut across the grain of one piece of wood and another piece of wood is fitted into the

5 Welding Shop 6 Hrs

- Welding is a process of joining two or more metal pieces by applying heat, pressure, or both, and with or without filler material.
- Welding can be classified into two main types: fusion welding and solid-state welding.

- Fusion welding is a type of welding where the base metal is melted and fused together, with or without filler metal. Examples of fusion welding are arc welding, gas welding, and resistance welding.
- Solid-state welding is a type of welding where the base metal is not melted, but joined by applying pressure and/or heat, and sometimes with the help of an intermediate layer. Examples of solid-state welding are forge welding, friction welding, and ultrasonic welding.
- Welding can also be classified based on the source of heat, such as electric arc welding, oxy-fuel gas welding, laser welding, and electron beam welding.
- Welding can also be classified based on the position of the workpiece, such as flat welding, horizontal welding, vertical welding, and overhead welding.
- Welding can also be classified based on the mode of metal transfer, such as short-circuiting transfer, globular transfer, spray transfer, and pulsed transfer.
- Welding can also be classified based on the shielding method, such as shielded metal arc welding (SMAW), gas metal arc welding (GMAW), gas tungsten arc welding (GTAW), and flux-cored arc welding (FCAW).
- Welding can also be classified based on the type of joint, such as butt joint, lap joint, corner joint, tee joint, and edge joint.
- Welding can also be classified based on the type of weld, such as fillet weld, groove weld, plug weld, slot weld, and spot weld.
- Welding shop is a place where welding operations are performed, using various welding equipment, tools, and accessories.
- Welding shop safety is important to prevent injuries, fires, explosions, and health hazards from welding.
- Welding shop safety rules include wearing proper personal protective equipment (PPE), such as gloves, goggles, helmets, aprons, and boots, keeping the work area clean and well-ventilated, following the manufacturer's instructions for using and maintaining the welding equipment, avoiding contact with live electrical parts, using fire extinguishers and first aid kits in case of emergencies, and reporting any accidents or hazards to the supervisor.

Introduction to BI standards and reading of welding drawings. Practice of Making following operations Butt Joint Lap Joint TIG Welding MIG Welding

- BI standards are the British standards for welding and fabrication. They specify the requirements for materials, design, execution, testing and inspection of welded structures and components.
- Reading of welding drawings involves interpreting the symbols, dimensions, notes and specifications that indicate the type, location, size and quality of welds and joints in a drawing.
- A butt joint is a type of joint where two pieces of metal are joined along a single edge or plane. The joint can be welded from one or both sides,

depending on the thickness and accessibility of the metal pieces.

- A lap joint is a type of joint where two pieces of metal are overlapped and welded together. The joint can be single or double, depending on the number of welds applied to the overlapping area.
- TIG welding (tungsten inert gas welding) is a welding process that uses a non-consumable tungsten electrode and an inert gas (usually argon) to create an arc between the electrode and the metal. The arc melts the metal and forms a weld pool, which is protected from the atmosphere by the gas. A filler metal may or may not be used, depending on the application and the metal type.
- MIG welding (metal inert gas welding) is a welding process that uses a consumable wire electrode and an inert gas (usually argon or carbon dioxide) to create an arc between the electrode and the metal. The arc melts the electrode and the metal, forming a weld pool that is protected from the atmosphere by the gas. The wire electrode is continuously fed from a spool through a welding gun, which controls the current, voltage and gas flow.

6 Moulding and Casting Shop 6 Hrs

- Moulding and casting are processes of shaping materials by pouring a liquid or molten material into a mould, where it solidifies into the desired shape.
- Moulding and casting are used for making various objects such as sculptures, jewellery, tools, parts, and more.
- Moulding and casting require different materials and equipment depending on the type and complexity of the object to be made.

Moulding Materials and Methods

- A mould is a hollow container that holds the liquid or molten material and defines the shape of the object to be made.
- Moulds can be made of different materials such as plaster, clay, wax, silicone, rubber, metal, or wood.
- Moulds can be made by different methods such as carving, sculpting, modelling, impression, or casting.
- Moulds can be classified into two types: open moulds and closed moulds.
  - Open moulds are moulds that have only one side, such as a bowl or a tray. They are used for simple shapes that do not have undercuts or complex details.
  - Closed moulds are moulds that have two or more parts, such as a box or a shell. They are used for complex shapes that have undercuts or intricate details. They require a parting line and a release agent to separate the mould parts and the object.

Casting Materials and Methods

- Casting is the process of pouring a liquid or molten material into a mould, where it cools and hardens into the shape of the mould.
- Casting materials can be classified into two types: cold casting and hot casting.
  - Cold casting is casting with materials that are liquid at room temperature, such as plaster, resin, latex, or silicone. They are mixed with water, catalyst, or hardener and then poured into the mould. They cure and harden within a few minutes or hours.
  - Hot casting is casting with materials that are solid at room temperature, such as metal, wax, or glass. They are heated to a high temperature and then poured into the mould. They cool and solidify within a few seconds or minutes.
- Casting methods can vary depending on the type and properties of the casting material, such as viscosity, density, shrinkage, and melting point. Some common casting methods are:
  - Gravity casting: Casting by pouring the material into the mould by gravity. It is the simplest and most common method of casting. It is suitable for most materials and shapes.
  - Pressure casting: Casting by forcing the material into the mould by applying pressure. It is used for materials that have low viscosity or high density, such as metal or ceramic. It can produce finer details and smoother surfaces than gravity casting.
  - Vacuum casting: Casting by creating a vacuum in the mould and then filling it with the material. It is used for materials that have high viscosity or low density, such as resin or foam. It can eliminate air bubbles and defects in the object.
  - Centrifugal casting: Casting by rotating the mould and then pouring the material into it. It is used for materials that have high melting point or low fluidity, such as metal or glass. It can produce hollow or cylindrical objects with uniform thickness and density.

Introduction to Patterns, pattern allowances, ingredients of moulding sand and melting furnaces. Foundry tools and their purposes Demo of mould preparation and Aluminum casting Practice – Study and Preparation of mould for Plastic

- Patterns are the replicas of the desired castings that are used to shape the mould cavities. They are usually made of wood, metal, plastic or other materials.
- Pattern allowances are the modifications made to the pattern dimensions

to compensate for various factors such as shrinkage, machining, distortion and draft.

- Ingredients of moulding sand are the materials that are mixed with water and binder to form the moulding medium. They include silica sand, clay, coal dust, cereal, molasses, etc.
- Melting furnaces are the devices that are used to melt the metal for casting. They can be classified into two types: fuel-fired furnaces and electric furnaces. Fuel-fired furnaces use coal, coke, oil or gas as the fuel source, while electric furnaces use electric current or induction to heat the metal.
- Foundry tools are the instruments that are used to perform various operations in the foundry, such as moulding, melting, pouring, cleaning and finishing. They include rammers, trowels, sprues, gates, runners, risers, vents, ladles, skimmers, chisels, hammers, files, etc.
- Demo of mould preparation and aluminum casting is a practical exercise that involves the following steps:
  - Selecting a suitable pattern and applying the pattern allowances.
  - Preparing the moulding sand by mixing the ingredients and adding water and binder.
  - Making the mould cavity by placing the pattern in a moulding box and packing the sand around it.
  - Removing the pattern and making the sprue, gate, runner, riser and vent holes in the mould.
  - Melting the aluminum in a furnace and skimming the impurities from the surface.
  - Pouring the molten metal into the mould cavity through the sprue and gate.
  - Allowing the metal to solidify and cool in the mould.
  - Breaking the mould and removing the casting.
  - Cleaning and finishing the casting by removing the excess metal and sand, and filing or grinding the surface.
- Practice – Study and Preparation of mould for Plastic is a similar exercise that involves the following steps:
  - Selecting a suitable pattern and applying the pattern allowances.
  - Preparing the moulding sand by mixing the ingredients and adding water and binder.
  - Making the mould cavity by placing the pattern in a moulding box and packing the sand around it.
  - Removing the pattern and making the sprue, gate, runner, riser and vent holes in the mould.
  - Melting the plastic in a furnace or a heating chamber and stirring it to avoid bubbles.
  - Pouring the molten plastic into the mould cavity through the sprue and gate.
  - Allowing the plastic to solidify and cool in the mould.
  - Breaking the mould and removing the plastic part.
  - Cleaning and finishing the plastic part by removing the excess plastic

and sand, and filing or sanding the surface.

7 CNC Shop 6 Hrs

CNC stands for Computer Numerical Control. It is a process of using computers to control machines that can perform various tasks such as cutting, drilling, milling, turning, etc. CNC machines can produce precise and complex shapes from different materials such as metal, wood, plastic, etc.

Some of the topics that can be covered in a 6-hour CNC shop course are:

- Introduction to CNC machines: types, components, functions, advantages and disadvantages of CNC machines.
- CNC programming: basic concepts, codes, formats, commands, parameters, variables, expressions, etc. How to write, edit, simulate and run CNC programs using software such as G-code, M-code, CAD/CAM, etc.
- CNC operations: how to set up, operate, monitor and troubleshoot CNC machines. How to select, mount, align and change tools, workpieces, fixtures, etc. How to measure, inspect and verify the quality and accuracy of CNC products.
- CNC safety: how to follow safety rules and regulations, wear protective equipment, avoid hazards and risks, handle emergencies, etc. How to maintain and clean CNC machines and tools. How to dispose of waste and scrap materials properly.

Study of main features and working parts of CNC machine and accessories that can be used. Perform different operations on metal components using any CNC machines

- CNC stands for Computer Numerical Control, which is a process of using a computer-driven machine tool to produce a part out of solid material in a different shape.
- CNC machines can perform various operations on metal components, such as drilling, milling, turning, threading, tapping, boring, cutting, etc.
- CNC machines have the following main features and parts :
  - Input device: This is the device that is used to input part programs in a CNC machine. Part programs are a set of instructions that tell the machine how to move and what to do. Input devices can be punch tape readers, magnetic tape readers, or computers via RS-232-C communication.
  - Machine control unit (MCU): This is the heart of the CNC machine. It receives the part program from the input device and converts it into electrical signals that control the motion and speed of the machine tools. It also monitors the feedback system and displays the status of the machine on the display unit.
  - Machine tools: These are the components that actually perform the cutting or shaping of the material. A CNC machine tool always has

a sliding table and a spindle to control position and speed. The table can move along one or more axes (X, Y, Z) and the spindle can rotate at different speeds and angles. The machine tools can be attached with various cutting tools, such as drills, mills, lathes, etc.
  - Driving system: This is the system that provides power and motion to the machine tools. It consists of motors, gears, belts, pulleys, etc. The driving system is controlled by the MCU according to the part program.
  - Feedback system: This is the system that measures the actual position and speed of the machine tools and sends the information back to the MCU. It consists of sensors, encoders, transducers, etc. The feedback system ensures the accuracy and precision of the CNC machine by correcting any errors or deviations from the part program.
  - Display unit: This is the device that shows the current status and operation of the CNC machine. It can be a monitor, a keyboard, a mouse, etc. The display unit allows the operator to interact with the CNC machine and modify the part program if needed.
  - Bed: This is the base or foundation of the CNC machine. It supports the machine tools and the workpiece. It is usually made of cast iron or steel and has a rigid and stable structure.
  - Headstock: This is the part of the CNC machine that holds the spindle and the driving system. It is mounted on the bed and can be moved along the Z-axis. It also contains the coolant system, the lubrication system, and the chip removal system.
- CNC machines can use various accessories to enhance their performance and functionality. Some of the common accessories are:
  - Tool holders: These are the devices that hold the cutting tools on the spindle. They can be collets, chucks, end mill holders, etc. They allow the quick and easy change of cutting tools and ensure their alignment and stability.
  - Workholding devices: These are the devices that hold the workpiece on the table. They can be vises, clamps, fixtures, etc. They ensure the proper positioning and orientation of the workpiece and prevent it from moving or slipping during the machining process.
  - Coolant system: This is the system that provides a liquid or gas to cool down the cutting tools and the workpiece. It prevents overheating, friction, and wear and tear. It also improves the surface finish and the tool life. The coolant can be water, oil, air, etc.
  - Lubrication system: This is the system that provides a lubricant to reduce friction and wear between the moving parts of the CNC machine. It also prevents rust and corrosion. The lubricant can be oil, grease, etc.
  - Chip removal system: This is the system that removes the chips or shavings that are produced during the machining process. It prevents the accumulation of chips on the machine tools and the workpiece. It also improves the safety and cleanliness of the CNC machine. The

chip removal system can be a conveyor belt, a vacuum, a blower, etc.

8 To prepare a product using 3D printing 3 Hrs

3D printing is a process of creating a three-dimensional object from a digital model by depositing successive layers of material on top of each other. 3D printing can be used to create various products, such as prototypes, models, sculptures, jewelry, tools, etc.

To prepare a product using 3D printing, the following steps are required:

- **Design the product**: The first step is to design the product using a computer-aided design (CAD) software or a 3D scanner. The design should be compatible with the 3D printer and the material to be used. The design should also be optimized for the printing process, such as reducing the number of supports, avoiding overhangs, and ensuring sufficient wall thickness.
- **Slice the model**: The second step is to slice the model into thin layers using a slicing software. The slicing software converts the 3D model into a series of instructions for the 3D printer, such as the movement of the nozzle, the extrusion of the material, the temperature of the bed and the nozzle, etc. The slicing software also generates a preview of the print and an estimate of the printing time and material usage.
- **Print the product**: The third step is to print the product using a 3D printer. The 3D printer follows the instructions from the slicing software and deposits the material layer by layer on the print bed. The material can be plastic, metal, resin, ceramic, or other materials, depending on the type of the 3D printer and the desired properties of the product. The printing process can take from minutes to hours, depending on the size and complexity of the product.
- **Post-process the product**: The fourth step is to post-process the product after the printing is completed. The post-processing can include removing the supports, sanding, polishing, painting, gluing, or other finishing techniques, depending on the product and the material. The post-processing can improve the appearance, strength, and functionality of the product.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of Total 33 Hrs. I assume you are referring to the total hours of sleep deprivation that can cause hallucinations and cognitive impairment. Here are some points to know about this topic:

- Sleep deprivation is the condition of not getting enough sleep, either voluntarily or involuntarily. It can have negative effects on physical and mental health, such as impaired memory, concentration, mood, immune system, and metabolism.

- Sleep deprivation can also cause microsleeps, which are brief episodes of sleep that occur involuntarily during wakefulness. Microsleeps can impair alertness, performance, and safety, especially when driving or operating machinery.
- One of the most extreme effects of sleep deprivation is hallucinations, which are sensory perceptions that occur without any external stimulus. Hallucinations can affect any of the five senses, such as seeing, hearing, feeling, smelling, or tasting things that are not there.
- Hallucinations can occur after as little as 24 hours of sleep deprivation, but they become more frequent and intense after 33 hours of continuous wakefulness. This is because the brain tries to compensate for the lack of sleep by entering a state of REM (rapid eye movement) sleep while awake, which is normally associated with dreaming.
- Hallucinations can be frightening, confusing, or amusing, depending on the content and context of the perception. Some common types of hallucinations are:
  - Visual hallucinations: seeing shapes, colors, patterns, objects, people, or animals that are not there. For example, seeing spiders crawling on the wall, or seeing a deceased relative.
  - Auditory hallucinations: hearing sounds, voices, music, or noises that are not there. For example, hearing someone calling your name, or hearing a phone ringing.
  - Tactile hallucinations: feeling sensations, such as touch, pain, temperature, or movement, that are not there. For example, feeling bugs crawling on your skin, or feeling someone tapping your shoulder.
  - Olfactory hallucinations: smelling odors, such as smoke, flowers, or food, that are not there. For example, smelling smoke when there is no fire, or smelling rotten eggs.
  - Gustatory hallucinations: tasting flavors, such as sweet, sour, bitter, or salty, that are not there. For example, tasting metal in your mouth, or tasting lemon when drinking water.
- Hallucinations can be influenced by several factors, such as the level of sleep deprivation, the individual's personality, mood, expectations, beliefs, and cultural background. Some people may be more prone to hallucinations than others, depending on their genetic, psychological, and environmental factors.
- Hallucinations can be distinguished from illusions, which are misinterpretations of real stimuli. For example, mistaking a coat rack for a person, or hearing a song in the sound of a fan. Illusions can also occur due to sleep deprivation, but they are less severe and more easily corrected than hallucinations.
- Hallucinations can be treated by getting enough sleep, which usually re-

stores normal brain function and perception. However, some people may experience persistent or recurrent hallucinations, which may indicate an underlying mental disorder, such as schizophrenia, bipolar disorder, or dementia. In such cases, professional help and medication may be needed.

Course Outcome:

- A course outcome is a statement that describes what students should be able to do or demonstrate by the end of a course.
- Course outcomes are usually derived from the course objectives, which are the broad goals or purposes of the course.
- Course outcomes are more specific and measurable than course objectives, and they should align with the course content, activities, and assessments.
- Course outcomes should be written using action verbs that indicate the level of cognitive, affective, or psychomotor skills that students are expected to achieve.
- Course outcomes should be clear, concise, and observable, and they should reflect the intended learning outcomes of the course.
- Course outcomes should be communicated to the students at the beginning of the course, and they should be used to guide the instruction and evaluation throughout the course.
- Course outcomes should be reviewed and revised periodically to ensure their relevance, validity, and effectiveness.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some content in markdown format that you can use for your study material.

Engineering Materials and Manufacturing Processes

CO1: Use various engineering materials, tools, machines and measuring equipments.

- Engineering materials are the substances that are used to create or manufacture products, structures, machines, and systems. They can be classified into metals, ceramics, polymers, composites, and others based on their properties and applications.
- Tools are the devices or instruments that are used to perform a specific task or operation. They can be hand tools, such as hammers, screwdrivers, wrenches, etc., or power tools, such as drills, saws, grinders, etc.
- Machines are the devices or systems that use energy to perform work or produce output. They can be simple machines, such as levers, pulleys, wheels, etc., or complex machines, such as engines, pumps, turbines, etc.
- Measuring equipments are the devices or instruments that are used to measure or quantify physical quantities, such as length, mass, force, tem-

perature, pressure, etc. They can be analog or digital, such as rulers, scales, calipers, thermometers, gauges, etc.

CO2: Perform machine operations in lathe and CNC machine.

- A lathe is a machine tool that rotates a workpiece about an axis of rotation and performs various operations, such as cutting, drilling, turning, facing, threading, etc., by using a single-point cutting tool or a multipoint cutting tool.
- A CNC machine is a computer-controlled machine tool that can perform various operations, such as milling, drilling, tapping, boring, etc., by using a rotating multipoint cutting tool or a non-rotating tool, such as a laser, plasma, water jet, etc.
- To perform machine operations in lathe and CNC machine, one needs to follow the steps of:
  - Selecting the appropriate material, tool, and machine for the operation.
  - Setting up the workpiece and the tool on the machine according to the specifications and instructions.
  - Programming the machine with the desired parameters and commands, such as speed, feed, depth of cut, coordinates, etc.
  - Running the machine and monitoring the operation for quality and safety.
  - Finishing the operation and removing the workpiece and the tool from the machine.

CO3: Perform manufacturing operations on components in fitting and carpentry shop.

- Fitting is the process of assembling or joining two or more components by using various methods, such as filing, drilling, tapping, riveting, etc.
- Carpentry is the process of making or repairing wooden structures or objects by using various methods, such as sawing, planing, chiseling, nailing, etc.
- To perform manufacturing operations on components in fitting and carpentry shop, one needs to follow the steps of:
  - Selecting the appropriate material, tool, and method for the operation.
  - Marking the dimensions and locations of the components on the material according to the specifications and instructions.
  - Cutting, shaping, and finishing the components by using the tool and the method.
  - Assembling or joining the components by using the appropriate fasteners or adhesives.

CO4: Perform operations in welding, moulding, casting and gas cutting.

- Welding is the process of joining two or more metal parts by applying heat, pressure, or both, and causing the parts to fuse or intermix.
- Moulding is the process of shaping a material, such as metal, plastic, or rubber, by using a hollow container or a pattern, called a mould, that has the desired shape and size.
- Casting is the process of pouring a molten material, such as metal, into a mould and allowing it to solidify and form the desired shape and size.
- Gas cutting is the process of cutting a metal part by using a flame that is produced by burning a fuel gas, such as acetylene, propane, etc., and an oxidizer gas, such as oxygen, air, etc.
- To perform operations in welding, moulding, casting and gas cutting, one needs to follow the steps of:
  - Selecting the appropriate material, tool, and method for the operation.
  - Preparing the parts or the mould for the operation by cleaning, heating, coating, etc.
  - Performing the operation by using the tool and the method according to the specifications and instructions.
  - Finishing the operation by cooling, removing, cleaning, inspecting, etc.

CO5: Fabricate a job by 3D printing manufacturing technique.

- 3D printing is a manufacturing technique that creates a three-dimensional object

Reference Books:

- Reference books are books that contain factual information on various topics, such as dictionaries, encyclopedias, atlases, directories, handbooks, manuals, etc.
- Reference books are usually consulted for specific information, rather than read cover to cover.
- Reference books are often arranged alphabetically, chronologically, geographically, or by subject, depending on the type and purpose of the book.
- Reference books are usually kept in a separate section of a library, called the reference collection, and are not available for borrowing.
- Reference books are useful for finding quick facts, definitions, statistics, maps, illustrations, bibliographies, and other sources of information on a topic.
- Reference books are updated regularly to reflect the latest information

and developments in various fields of knowledge.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is a summary of the topic you requested:

1. **Workshop Practice, H S Bawa, McGraw Hill**

   - This book covers the basic concepts and skills of workshop practice for engineering students.
   - It includes topics such as safety, tools, measuring instruments, materials, machining processes, welding, fitting, carpentry, sheet metal work, plumbing, and electrical work.
   - It also provides practical exercises, projects, and multiple choice questions for self-assessment.
   - The book is divided into 16 chapters, each covering a specific aspect of workshop practice.
   - The chapters are:
     - Chapter 1: Introduction to Workshop Practice
     - Chapter 2: Safety in Workshop
     - Chapter 3: Tools and Measuring Instruments
     - Chapter 4: Engineering Materials
     - Chapter 5: Bench Work and Fitting
     - Chapter 6: Carpentry and Pattern Making
     - Chapter 7: Sheet Metal Work
     - Chapter 8: Smithy and Forging
     - Chapter 9: Welding
     - Chapter 10: Foundry
     - Chapter 11: Machine Tools and Machining Operations
     - Chapter 12: Lathe Machine
     - Chapter 13: Drilling Machine
     - Chapter 14: Milling Machine
     - Chapter 15: Shaping, Planning and Slotting Machines
     - Chapter 16: Plumbing and Electrical Work
   - The book is written in a simple and concise manner, with diagrams, tables, and photographs to illustrate the concepts and procedures.
   - The book is suitable for diploma and degree level students of mechanical, civil, electrical, and production engineering.

Mechanical Workshop Practice, K C John, PHI

- Mechanical Workshop Practice is a book by K C John, published by PHI Learning in 2010.

- The book is intended for students of mechanical engineering and related disciplines who need to learn the basic skills and techniques of workshop practice.
- The book covers the following topics:
  - Introduction to workshop practice, safety, care and precaution in workshop
  - Primary and secondary shaping processes, such as casting, forging, sheet metal work, machining, etc.
  - Metal joining methods, such as welding, brazing, soldering, riveting, etc.
  - Surface finishing and heat treatment, such as grinding, polishing, hardening, tempering, etc.
  - Measurement and inspection, such as linear and angular measurements, gauges, comparators, etc.
  - Machine tools, such as lathe, drilling machine, milling machine, etc.
  - Fitting and assembly, such as cutting, filing, drilling, tapping, etc.
  - Carpentry and pattern making, such as wood working tools, joints, patterns, etc.
  - Smithy and forging, such as hand tools, forging operations, defects, etc.
  - Foundry and casting, such as moulding materials, sand testing, casting defects, etc.
- The book is written in a simple and lucid style, with numerous illustrations, examples, and exercises.
- The book aims to provide a comprehensive and practical knowledge of workshop practice to the students and enable them to apply the skills in various engineering projects.

Workshop Practice Vol 1, and Vol 2, by Hazra-Choudhary , Media promoters and Publications

- Workshop Practice is a subject that covers the basic skills and knowledge of various workshop processes, such as fitting, sheet metal, soldering, welding, carpentry, foundry, smithy, etc.
- Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications are two books that provide a comprehensive and practical guide to workshop practice for engineering students and professionals.
- Workshop Practice Vol 1 covers the topics of manufacturing processes, such as casting, forging, welding, machining, metal cutting, metal forming, etc. It also includes chapters on metrology, engineering materials, heat treatment, and surface finishing.

- Workshop Practice Vol 2 covers the topics of machine tools, such as lathe, drilling, milling, shaping, planing, slotting, grinding, etc. It also includes chapters on tool engineering, jigs and fixtures, press tools, dies and moulds, and CNC machines.
- The books are written in a simple and lucid language, with numerous illustrations, diagrams, tables, and examples. The books also contain exercises, review questions, and objective type questions at the end of each chapter.
- The books are suitable for diploma, degree, and AMIE courses in mechanical, production, automobile, and industrial engineering. They are also useful for ITI and vocational courses, as well as for practicing engineers and technicians.

CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication

- This is a text-book that explains the fundamentals of NC/CNC machine tools, operations and part programming which form essential portion of course on Computer Aided Manufacturing (CAM).
- This book also covers advanced topics such as Macro programming, DNC and Computer Aided Part Programming (CAPP) in detail.
- The book is divided into 12 chapters, each covering a specific aspect of CNC technology and its applications.
- The book is written in a simple and easy to understand language, with numerous examples, diagrams, tables and exercises.
- The book is suitable for undergraduate and postgraduate students of mechanical engineering, production engineering, industrial engineering and other related disciplines.
- The book is also useful for practicing engineers, technicians and professionals who are involved in CNC machining and programming.

Chapter-wise summary of the book:

- Chapter 1: Introduction to NC/CNC Technology
  - This chapter introduces the basic concepts and definitions of NC/CNC technology, such as NC/CNC machine tools, NC/CNC systems, NC/CNC codes, NC/CNC modes, NC/CNC axes, NC/CNC coordinates, NC/CNC motions, NC/CNC interpolation, NC/CNC accuracy and NC/CNC advantages and disadvantages.
- Chapter 2: NC/CNC Machine Tools and Systems
  - This chapter describes the various types and components of NC/CNC machine tools, such as NC/CNC lathes, NC/CNC milling machines, NC/CNC drilling machines, NC/CNC grinding machines, NC/CNC

EDM machines, NC/CNC robots, etc. It also explains the functions and features of NC/CNC systems, such as NC/CNC controllers, NC/CNC drives, NC/CNC feedback devices, NC/CNC memory devices, NC/CNC display devices, NC/CNC input/output devices, etc.

- Chapter 3: NC/CNC Operations and Tooling
  - This chapter discusses the various types and parameters of NC/CNC operations, such as NC/CNC turning, NC/CNC facing, NC/CNC grooving, NC/CNC threading, NC/CNC boring, NC/CNC milling, NC/CNC drilling, NC/CNC tapping, NC/CNC reaming, NC/CNC contouring, NC/CNC pocketing, NC/CNC engraving, etc. It also explains the selection and design of NC/CNC tools and tool holders, such as NC/CNC single point tools, NC/CNC multipoint tools, NC/CNC form tools, NC/CNC tool materials, NC/CNC tool geometry, NC/CNC tool wear, NC/CNC tool life, NC/CNC tool compensation, etc.
- Chapter 4: NC/CNC Part Programming
  - This chapter introduces the concept and methodology of NC/CNC part programming, such as NC/CNC program format, NC/CNC program structure, NC/CNC program elements, NC/CNC program planning, NC/CNC program verification, NC/CNC program documentation, etc. It also covers the various types and methods of NC/CNC part programming, such as NC/CNC manual part programming, NC/CNC conversational part programming, NC/CNC graphical part programming, NC/CNC parametric part programming, NC/CNC adaptive part programming, etc.
- Chapter 5: NC/CNC Manual Part Programming
  - This chapter explains the principles and techniques of NC/CNC manual part programming, such as NC/CNC word address format, NC/CNC preparatory functions, NC/CNC miscellaneous functions, NC/CNC tool functions, NC/CNC motion functions, NC/CNC coordinate functions, NC/CNC canned cycles, NC/CNC subprograms, NC/CNC macros, NC/CNC expressions, NC/CNC variables, NC/CNC conditional statements, NC/CNC loops, NC/CNC arithmetic operations, NC/CNC logical operations, etc. It also provides several examples of NC/CNC manual part programs for different types of NC/CNC machine tools and operations.
- Chapter 6: NC/CNC Conversational Part Programming
  - This chapter describes the features and advantages of NC/CNC conversational part programming, such as NC/CNC menu-driven programming, NC/CNC interactive programming, NC/CNC graphical programming, NC/CNC simulation programming, NC/CNC editing

programming, etc. It also illustrates the use of NC/CNC conversational part programming for various types of NC/CNC machine tools and operations, such as NC/CNC turning, NC/CNC milling, NC/CNC drilling, NC/CNC contouring, etc.

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